Name: _____

TOPIC TEST 1

Chapter 1: Equations and polynomials

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 6 short answer questions (32 marks)
- Part C: 3 analysis questions (23 marks)

Part A

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 One or more errors have been made in the solution to this linear equation. State where the error(s) occurred.

$$\frac{x-1}{3} - \frac{5x-7}{4} = 2 - x$$

$$4(x-1) - 3(5x-7) = 2 - x \quad \text{Line 1}$$

$$4x - 4 - 15x - 21 = 2 - x \quad \text{Line 2}$$

$$-11x - 25 = 2 - x \quad \text{Line 3}$$

$$-10x = 27$$

$$x = -2.7$$

A Line 1 only.

B Line 2 only.

C Line 3 only.

D Line 1 and Line 2.

E Line 1, Line 2 and Line 3.

[1 mark]

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2 Solve the equation $2x^2 + 7x - 3 = 0$, for x correct to 2 decimal places.

A $x = -3.89$ and 0.39

B $x = -0.5$ and -3

C $x = -0.77$ and -7.72

D $x = 0.39$ and 3.89

E $x = 0.77$ and 7.72

[1 mark]

3 Given that y is inversely proportional to the square of x, and that $y = 8$ when $x = 3$, find the value of x when $y = 128$.

A $\frac{9}{16}$

B $\frac{3}{4}$

C $\frac{16}{9}$

D $\frac{4}{3}$

E 12

[1 mark]

4 Solve the equations $8a + b = -12$ and $4a - 3b = -20$ simultaneously.

A $a = 2, b = -28$

B $a = -3.5, b = 16$

C $a = -2.\overline{6},\ b = -3.\overline{1}$

D $a = 2, b = -28$

E $a = -2, b = 4$

[1 mark]

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5 Factorise $x^3 - 5x^2 - 12x + 36$.

A $(x + 2)(x - 3)(x + 6)$

B $(x - 2)(x - 3)(x + 6)$

C $(x - 2)(x + 3)(x - 6)$

D $(x + 2)(x - 3)(x - 6)$

E $(x - 2)(x - 3)(x - 6)$

[1 mark]

Part B

6 short answer questions
32 marks
Show your working where appropriate.

1 Solve the following equations.

a $2 + 5(m - 1) = 5m - (m - 15)$

b $\frac{p + 1}{2} - \frac{p + 2}{3} = 7$

[5 marks]

2 Solve the following equations.

a $(x - 4)(2x + 13) = 0$

b $x^2 + 6x - 12 = 0$ (by completing the square)

c $2x^2 - 3x - 5 = 0$ (by factorisation)

d $x^2 + 6x - 1 = 0$ (using the quadratic formula)

[10 marks]

3 Solve the equations $3x - y + 5 = 0$ and $y = 5x - 11$.

[4 marks]

4 The amount of fuel used by a car is directly proportional to the distance travelled. On a trip of 810 km, a family car used 60 litres of fuel. How far could the car travel on 76 litres of fuel?

[3 marks]

5 Given that $P(x) = 5x^3 + x^2 - 7x + 4$, find:

a the degree of the polynomial _____

b the leading term _____

c the constant _____

d the coefficients _____

e the value of $P(-2)$.

[5 marks]

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6 Divide the polynomial $P(x) = 5x^3 + x^2 - 7x + 4$ by $x + 2$ and hence write $P(x)$ in the form $P(x) = (x + 2)Q(x) + R$.

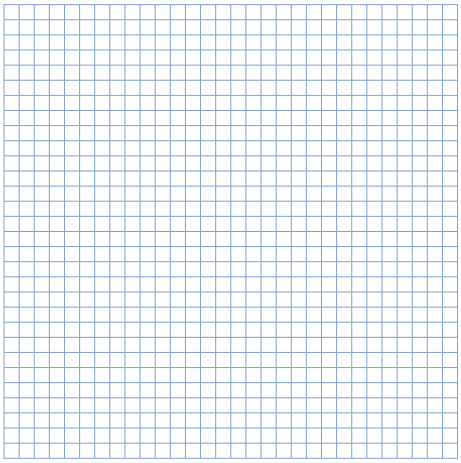
[5 marks]

Part C

3 analysis questions
23 marks
Show your working where appropriate.

1 A ski tour operator offers complete packages from Melbourne to the Victorian snowfields. The tours may be from 3 to 12 days in length and they cost $700 for the 3-day tour and $160 for each additional day.

a Draw a graph showing the cost of the tours provided by this operator.

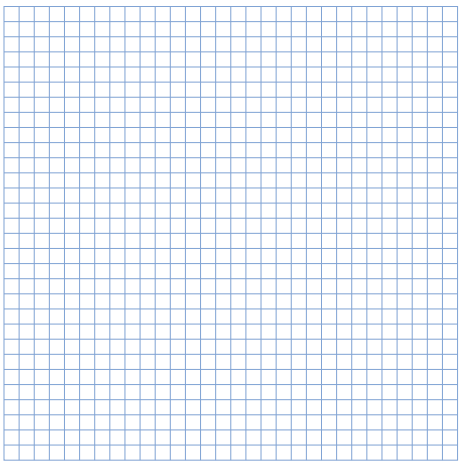


b Find a relationship between the number of days and the cost of the tours.

Another operator runs trips of any duration from 1 to 20 days with a flat rate of $190 per day.

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c Draw a graph of the cost of the tours provided by this operator.



d Find a relationship between the number of days and the cost of the tours.

e State the circumstances under which each operator would be cheaper to use.

[12 marks]

2 Robyn and Jake go on a hiking trip. Robyn starts off carrying the pack with the tent and sleeping bags, and Jake starts with the one containing the food and cooking gear. After they have gone 15 km, they swap packs for a change and discover that Jake's original pack is twice as heavy. They transfer 6 kg of food from Robyn's new pack to Jake's new pack, so that when they have completed the final 25 km of the hike, the total product of weight and distance will be the same for each person. Write an equation to find the original weight of each pack, then solve it to find the original weights.

[4 marks]

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3 Solve $3x^3 - 12x^2 + 3x + 18 = 0$

[7 marks]

Total marks: 60

Answers

Part A **[1 mark each]**

1 Errors in Lines 1 and 2 are shown in red.

$$\frac{x-1}{3}-\frac{5x-7}{4}=2-x$$

$4(x-1)-3(5x-7)=12(2-x)$ Line 1

$4x-4-15x+21=2-x$ Line 2

$-11x-25=2-x$ Line 3

$-10x=27$

$x=-2.7$

$\therefore$ **D** **[1 mark]**

2 $2x^2+7x-3=0$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$x=\frac{-7\pm\sqrt{7^2-4\times2\times(-3)}}{2\times2}$$

$$=\frac{-7\pm\sqrt{73}}{4}$$

$$=\frac{-7-\sqrt{73}}{4},\frac{-7+\sqrt{73}}{4}$$

$=-3.88600\ldots, 0.38600\ldots$

$\approx -3.89, 0.39$ correct to 2 decimal places

$\therefore$ **A** **[1 mark]**

3 $y=\frac{k}{x^2}$

$8=\frac{k}{3^2}$

$8=\frac{k}{9}$

$k=72$

$\therefore y=\frac{72}{x^2}$

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When $y = 128$

$$128 = \frac{72}{x^2}$$
$$128x^2 = 72$$
$$x^2 = \frac{72}{128}$$
$$x^2 = \frac{9}{16}$$
$$x = \frac{3}{4}$$

$\therefore$ **B** **[1 mark]**

4

$$8a + b = -12 \quad [1]$$
$$4a - 3b = -20 \quad [2]$$

$[1] \times 3 \quad 24a + 3b = -36 \quad [1a]$

$[1a] + [2] \quad 28a = -56$

$$a = \frac{-56}{28}$$
$$= -2$$

Substitute $a = -2$ into [1].

$8 \times (-2) + b = -12$

$-16 + b = -12$

$b = 4$

$\therefore$ Solution is $a = -2, b = 4$

$\therefore$ **E** **[1 mark]**

5 $x^3 - 5x^2 - 12x + 36$

$P(2) = 2^3 - 5(2)^2 - 12(2) + 36$

$= 0$

$\therefore (x - 2)$ is a factor

$$\begin{array}{r} x^2 - 3x - 18 \\ (x-2)\overline{)x^3 - 5x^2 - 12x + 36} \\ \underline{x^3 - 2x^2} \\ -3x^2 - 12x \\ \underline{-3x^2 + 6x} \\ -18x + 36 \\ \underline{-18x + 36} \\ 0 \end{array}$$

$\therefore x^3 - 5x^2 - 12x + 36 = (x - 2)(x^2 - 3x - 18)$

$= (x - 2)(x + 3)(x - 6)$

$\therefore$ C **[1 mark]**

Part B

1 a $2 + 5(m - 1) = 5m - (m - 15)$

$2 + 5m - 5 = 5m - m + 15$

$5m - 3 = 4m + 15$ **[1 mark]**

$m = 18$ **[1 mark]**

b $\frac{p + 1}{2} - \frac{p + 2}{3} = 7$ **[1 mark]**

$\frac{{}^{3}\cancel{6}(p+1)}{\cancel{2}_{1}} - \frac{{}^{2}\cancel{6}(p+2)}{\cancel{3}_{1}} = 6 \times 7$

$3(p + 1) - 2(p + 2) = 42$

$3p + 3 - 2p - 4 = 42$ **[1 mark]**

$p - 1 = 42$

$p = 43$ **[1 mark]**

2 a $(x - 4)(2x + 13) = 0$

$x - 4 = 0$ or $2x + 13 = 0$

$x = 4$ or $2x = -13$

$x = -6.5$

Solution: $x = -6.5, 4$ **[1 mark]**

b $x^2 + 6x - 12 = 0$

$x^2 + 6x = 12$

$x^2 + 6x + 9 = 12 + 9$ **[1 mark]**

$(x + 3)^2 = 21$ **[1 mark]**

$x + 3 = \pm\sqrt{21}$

$x = -3 \pm \sqrt{21}$ **[1 mark]**

c $2x^2 - 3x - 5 = 0$ $\begin{matrix}\times 10\\ +-3\end{matrix}$

$\frac{(x+1)}{\cancel{(2x + 2)}(2x - 5)}{\cancel{2}_{1}} = 0$ **[1 mark]**

$(x + 1)(2x - 5) = 0$

$x + 1 = 0$ or $2x - 5 = 0$ **[2 marks, 1 mark for each solution]**

$x = -1$ or $2x = 5$

$x = 2.5$

Solution: $x = -1, 2.5$

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d $x^2 + 6x - 1 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$ **[1 mark]**

$$x = \frac{-6 \pm \sqrt{6^2 - 4 \times 1 \times -1}}{2 \times 1}$$ **[1 mark]**

$$= \frac{-6 \pm \sqrt{40}}{2}$$ **[1 mark]**

$$= \frac{-6 \pm 2\sqrt{10}}{2}$$

$$= -3 \pm \sqrt{10}$$ **[1 mark]**

3 $3x - y + 5 = 0$ [1]

$y = 5x - 11$ [2]

Substitute [2] into [1].

$3x - (5x - 11) + 5 = 0$ **[1 mark]**

$3x - 5x + 11 + 5 = 0$

$-2x + 16 = 0$

$16 = 2x$

$x = 8$ **[1 mark]**

Substitute $x = 8$ into [2]. **[1 mark]**

$y = 5 \times 8 - 11$

$y = 29$ **[1 mark]**

Solution: $x = 8, y = 29$

4 The general equation for the amount of fuel used is: $f = kd$ **[1 mark]**

When $f = 60, d = 810$

$60 = 810k$

$$k = \frac{60}{810}$$

$$k = \frac{2}{27}$$ **[1 mark]**

$$\therefore f = \frac{2}{27}d$$

When $f = 76, d = ?$

$$76 = \frac{2}{27}d$$

$$d = 76 \times \frac{27}{2}$$

$d = 1026$ **[1 mark]**

The car can travel 1026 km on 76 litres of fuel.

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5 a The highest power of the polynomial is 3. **[1 mark]**

b The leading term contains the highest power, so it is $5x^3$. **[1 mark]**

c The constant has no pronumeral, i.e., the coefficient of x^0, so it is 4. **[1 mark]**

d 5, 1, −7, 4

The numbers in front of the x terms in descending order of power. **[1 mark]**

e $P(x) = 5x^3 + x^2 - 7x + 4$

$$\begin{aligned} P(-2) &= 5(-2)^3 + (-2)^2 - 7(-2) + 4 \\ &= 5 \times (-8) + 4 + 14 + 4 \\ &= -18 \end{aligned}$$ **[1 mark]**

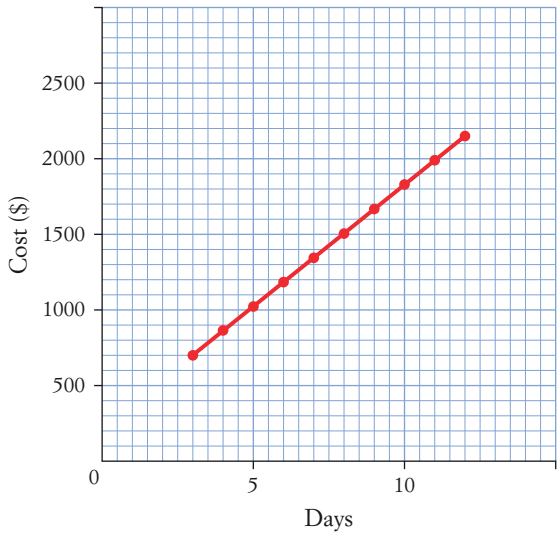
6 $P(x) = 5x^3 + x^2 - 7x + 4$

$$\begin{array}{r} 5x^2 - 9x + 11 \\ (x+2)\overline{)5x^3 + x^2 - 7x + 4} \\ \underline{5x^3 + 10x^2} \\ -9x^2 - 7x \\ \underline{-9x^2 - 18x} \\ 11x + 4 \\ \underline{11x + 22} \\ -18 \end{array}$$

$\therefore 5x^3 + x^2 - 7x + 4 = (x+2)(5x^2 - 9x + 11) - 18$ **[1 mark]**

Part C

1 a



[2 marks for graph —1 for axes and labelling, 1 for correct linear graph]

Days	3	4	5	6	7	8	9	10	11	12
Cost	700	860	1020	1180	1340	1500	1660	1820	1980	2140

[1 mark]

b Cost of 3 days at \$160 per day = \$480

But 3 days cost \$700, i.e., \$480 + \$220 **[1 mark]**

∴ Cost of 3 days = \$160 × 3 + \$220

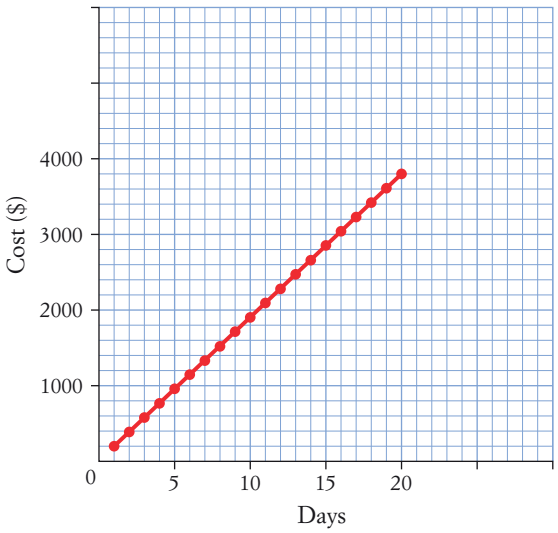
Cost of tour = $160d + 220$ **[1 mark]**

c

Days	1	2	3	4	5	6	7	8	9	10
Cost	190	380	570	760	950	1140	1330	1520	1710	1900

Days	11	12	13	14	15	16	17	18	19	20
Cost	2090	2280	2470	2660	2850	3040	3230	3420	3610	3800

[1 mark]



[2 marks for graph —1 for axes and labelling, 1 for correct linear graph]

d The cost is \$190 per day, i.e., Cost = \$190 × number of days **[1 mark]**

$C = 190d$ **[1 mark]**

e Looking at the two tables Day 8 is the first day that the second tour becomes more expensive than the first tour. **[1 mark]**

For tours between 3 and 7 days, the second operator is cheaper. For tours between 8 and 12 days, the first operator is cheaper. **[1 mark]**

2 $15s + 25(2s - 6) = 15 \times 2s + 25(s + 6)$, where s is the weight of the pack that was originally lighter; 30 kg and 60 kg

$15s + 25(2s - 6) = 15 \times 2s + 25(s + 6)$ **[2 marks —1 for each side of the equation]**

$$15s + 50s - 150 = 30s + 25s + 150$$

$$65s - 150 = 55s + 150$$

$$10s = 300$$

$$s = 30$$ **[1 mark]**

The heavier pack was twice the weight of the smaller pack.

∴ The original weight of the packs was 30 kg and 60 kg. **[1 mark]**

3 $3x^3 - 12x^2 + 3x + 18 = 0$

$3(x^3 - 4x^2 + x + 6) = 0$ **[1 mark]**

$P(-1) = (-1)^3 - 4(-1)^2 + (-1) + 6$

$= 0$

$\therefore (x + 1)$ is a factor. **[1 mark]**

[2 marks for division working out, minus 1 for any mistake]

$$
\begin{array}{r}
x^2 - 5x + 6 \\
(x+1)\overline{)\,x^3 - 4x^2 + x + 6} \\
\underline{x^3 + x^2} \\
-5x^2 + x \\
\underline{-5x^2 - 5x} \\
6x + 6 \\
\underline{6x + 6} \\
0
\end{array}
$$

$\therefore 3(x + 1)(x^2 - 5x + 6) = 0$ **[1 mark]**

$3(x + 1)(x - 2)(x - 3) = 0$ **[1 mark]**

$x + 1 = 0$ or $x - 2 = 0$ or $x - 3 = 0$ **[1 mark]**

$x = -1$ $x = 2$ $x = 3$

Solution: $x = -1, 2, 3$

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